Solid State and Devices

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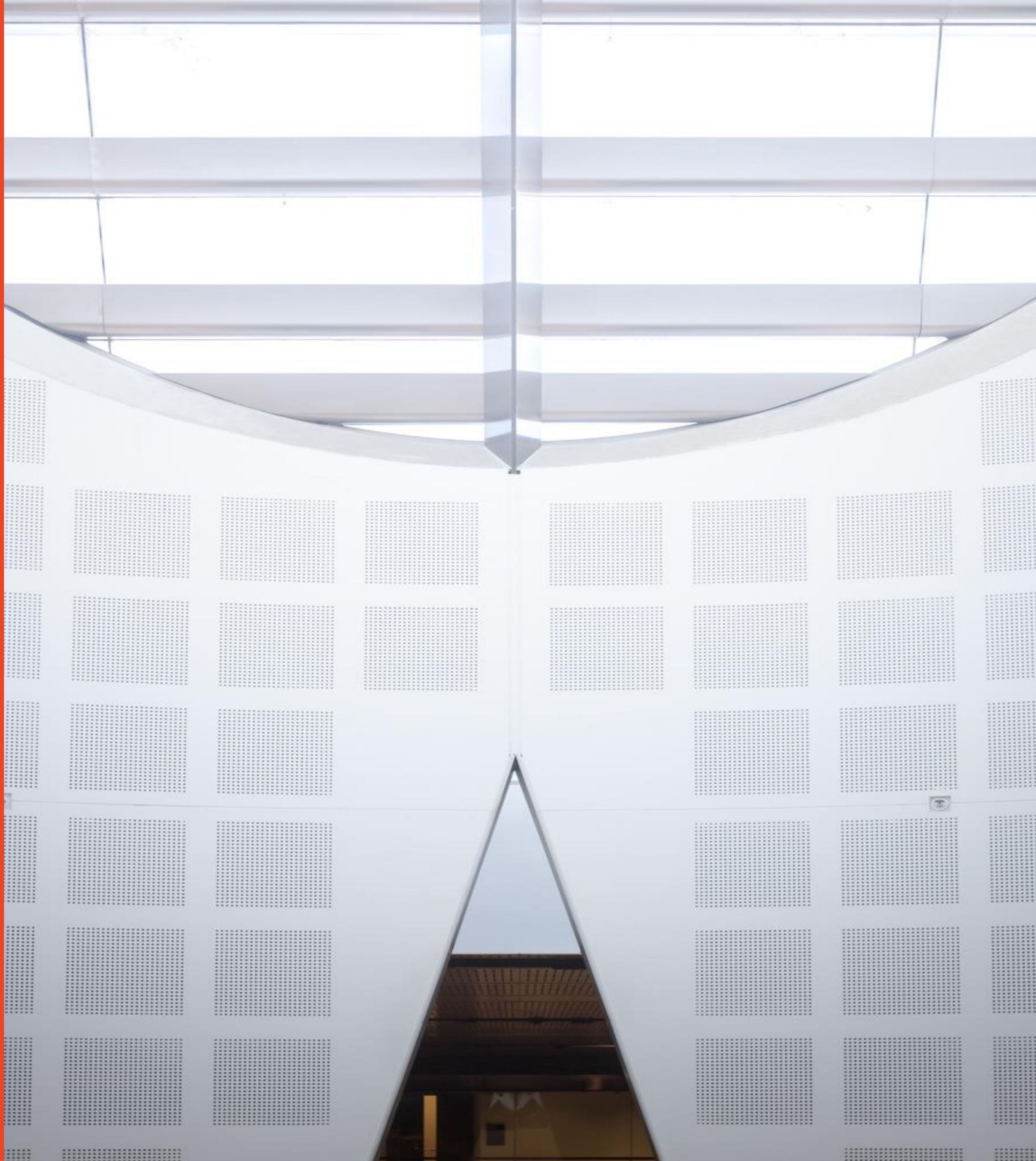


Intrinsic Semiconductors Electron concentration

- Number of thermally excited electrons in the conduction band equals the number of holes in the valence band:
- $n = p = n_i$,
- where n_i is the intrinsic carrier concentration.

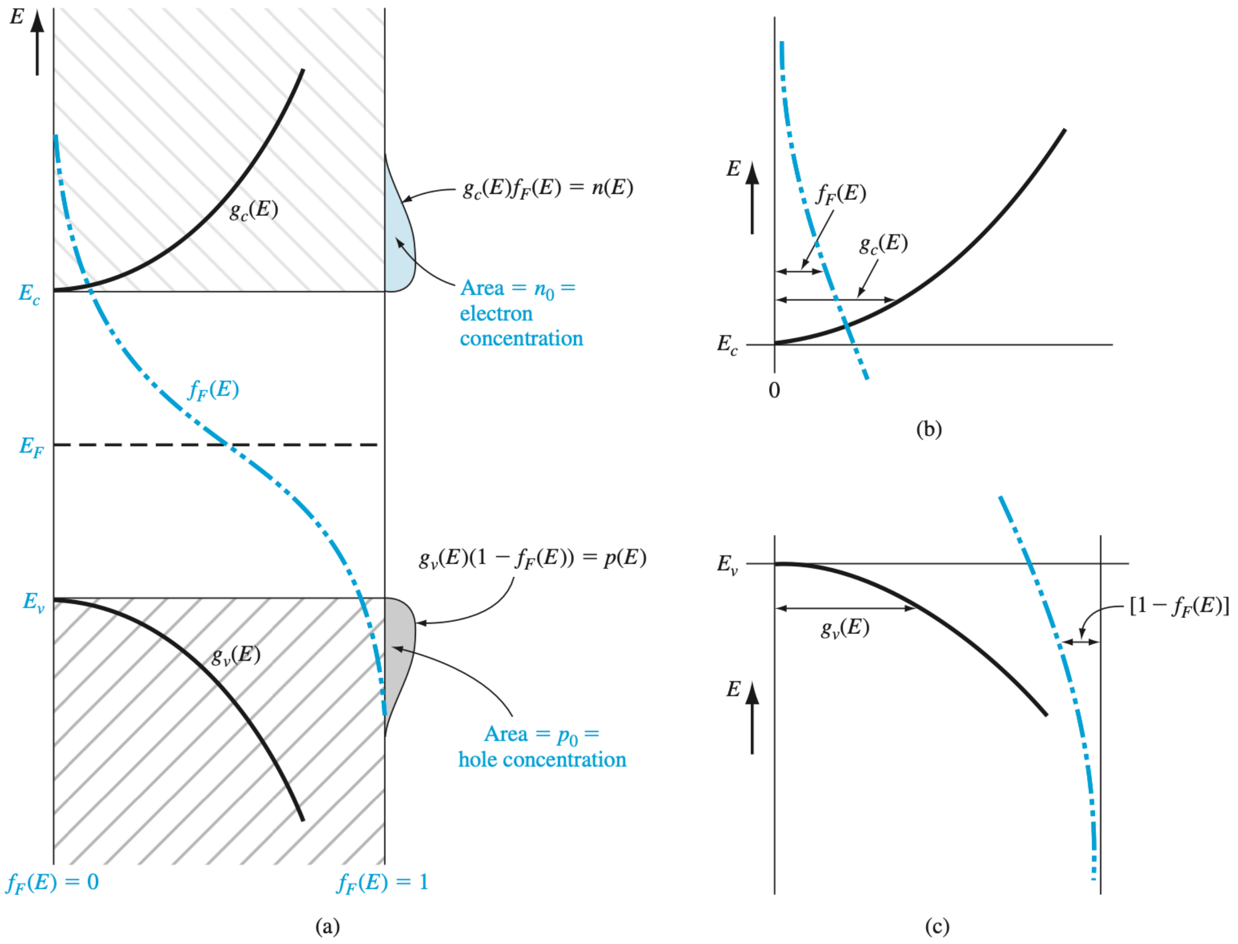


Figure 4.1 | (a) Density of states functions, Fermi–Dirac probability function, and areas representing electron and hole concentrations for the case when E_F is near the midgap energy; (b) expanded view near the conduction-band energy; and (c) expanded view near the valence-band energy.

- T>0K

$$n(E) = g_c(E) f_F(E)$$

$$n_0 = \int_{E_c}^{\infty} g_c(E) f_F(E)\, dE$$

Intrinsic Semiconductors Holes concentration

- If we assume effective masses of electron and hole are the same (band structure assumption)

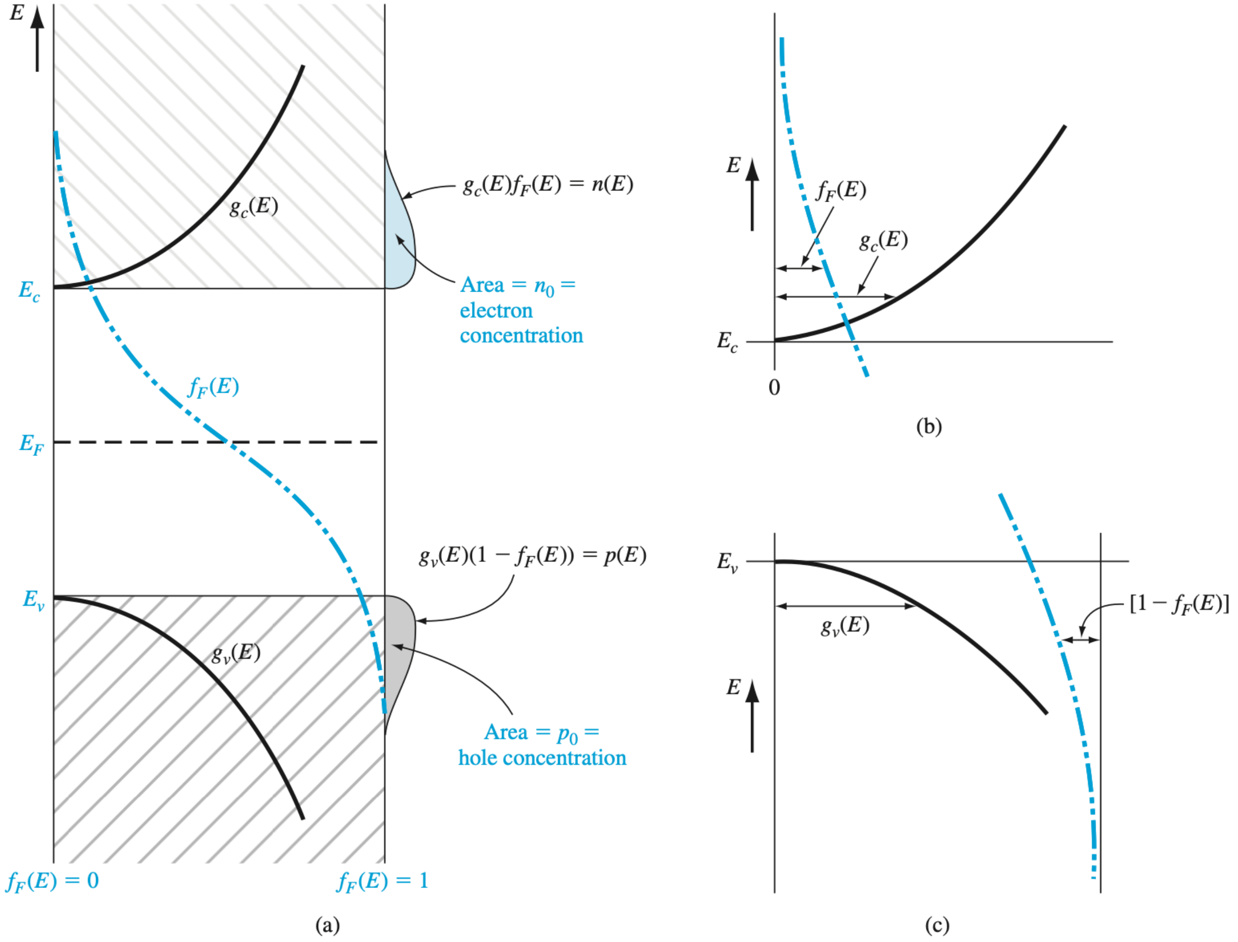
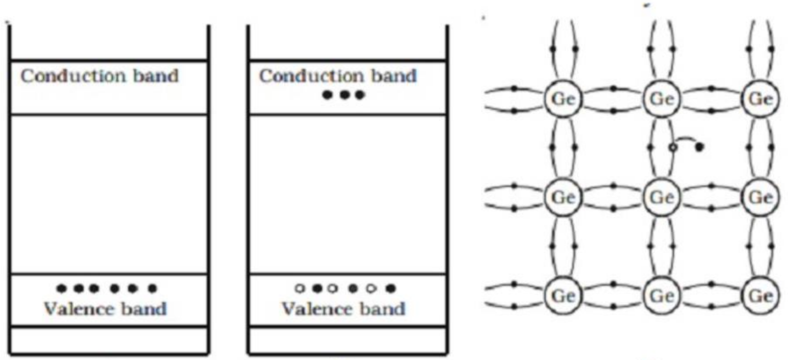


Figure 4.1 | (a) Density of states functions, Fermi–Dirac probability function, and areas representing electron and hole concentrations for the case when E_F is near the midgap energy; (b) expanded view near the conduction-band energy; and (c) expanded view near the valence-band energy.

- T>0K

$$p(E) = g_v(E)\,[1 - f_F(E)]$$

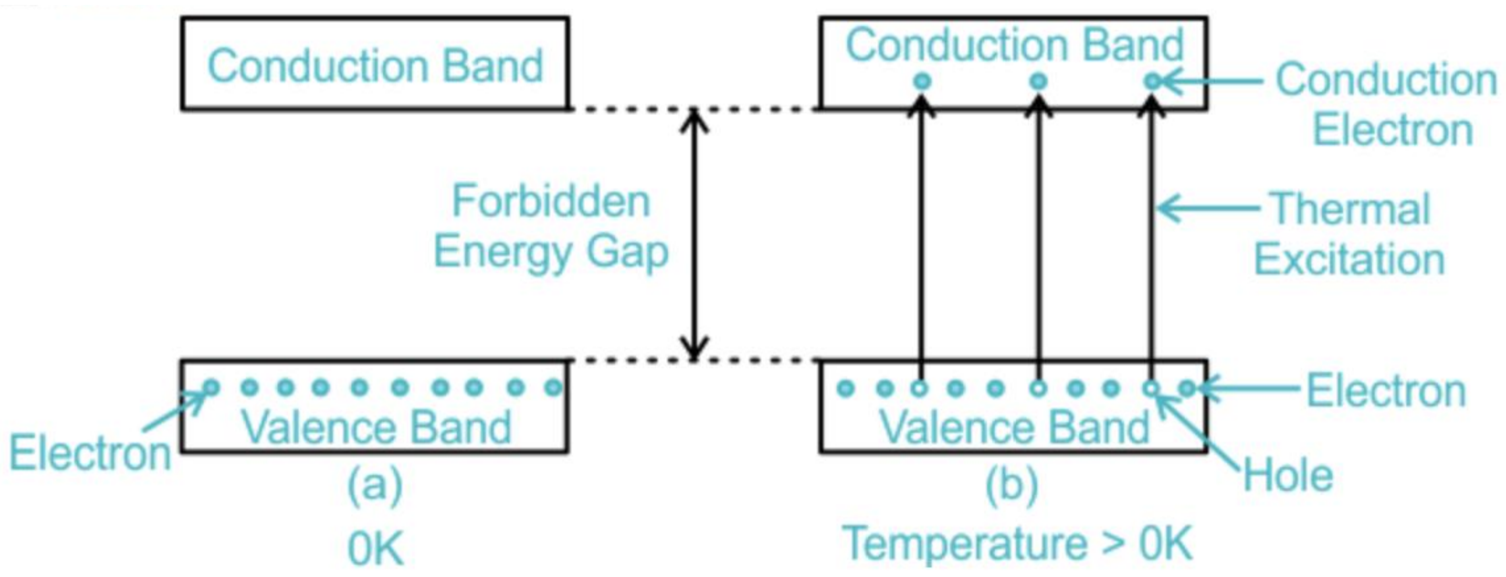


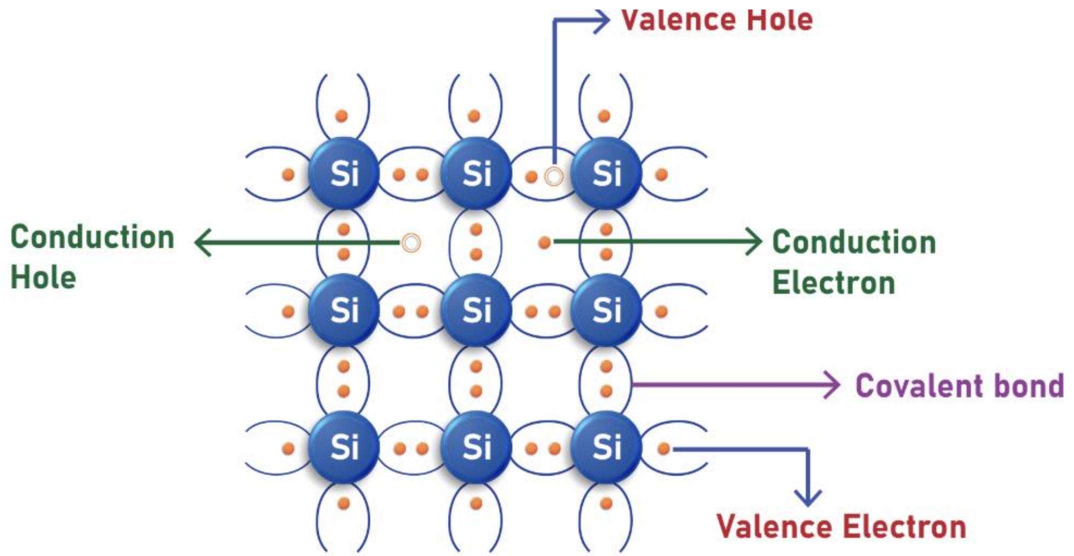
(a) (b) (c)

Fig a&b Electrons and holes in semiconductors

Fig c Formation of a hole

Intrinsic Semiconductors





Intrinsic Semiconductors Effective DOS

$$f_F(E) = \frac{1}{1 + \exp\frac{(E - E_F)}{kT}} \approx \exp\left[\frac{-(E - E_F)}{kT}\right]$$

$$g_c(E) = \frac{4\pi(2m_n^*)^{3/2}}{h^3}(E - E_c)^{1/2}.$$

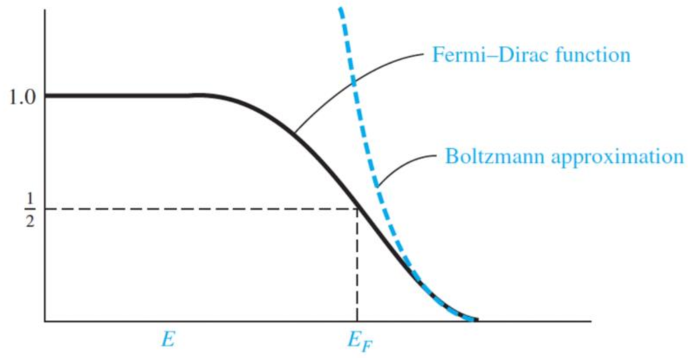
$$n_0 = \frac{4\pi(2m_n^*)^{3/2}}{h^3} \int_{E_c}^{\infty} \frac{(E - E_c)^{1/2} dE}{1 + \exp\left(\frac{E - E_F}{kT}\right)}.$$

- If (E_c – E_f) >> kT Fermi-Dirac distribution can be approximated to Boltzmann

If $E - E_F \gg kT$, we use $f_F(E) \approx \exp\left[-\frac{E - E_F}{kT}\right]$ and integrate:

$$n_0 = \frac{4\pi(2m_n^*)^{3/2}}{h^3} e^{\frac{E_F}{kT}} \int_{E_c}^{\infty} (E - E_c)^{1/2} e^{-\frac{E}{kT}} dE.$$

$$n_0 = 2\left(\frac{2\pi m_n^* kT}{h^2}\right)^{3/2} \exp\left(-\frac{E_c - E_F}{kT}\right).$$



$$\boxed{N_c \equiv 2\left(\frac{2\pi m_n^* kT}{h^2}\right)^{3/2} \quad \Rightarrow \quad n_0 = N_c \exp\left(-\frac{E_c - E_F}{kT}\right).}$$

Intrinsic Semiconductors carrier concentration

Effective density of states function in the conduction and valence band.

$$N_c \equiv 2\left(\frac{2\pi m_n^* kT}{h^2}\right)^{3/2} \quad \Rightarrow \quad n_0 = N_c \exp\left(-\frac{E_c - E_F}{kT}\right).$$

$$p_0 = N_v \exp\left(-\frac{E_F - E_v}{kT}\right), \qquad N_v = 2\left(\frac{2\pi m_p^* kT}{h^2}\right)^{3/2}.$$

For an intrinsic semiconductor, the number of electrons in the conduction band is equal to the number of holes in the valence band $n_i = p_i$.

$$n_i = \sqrt{N_c N_v}\exp\left(-\frac{E_g}{2kT}\right).$$

Conditions for Intrinsic Semiconductor at Equilibrium $n_i = p_i$.

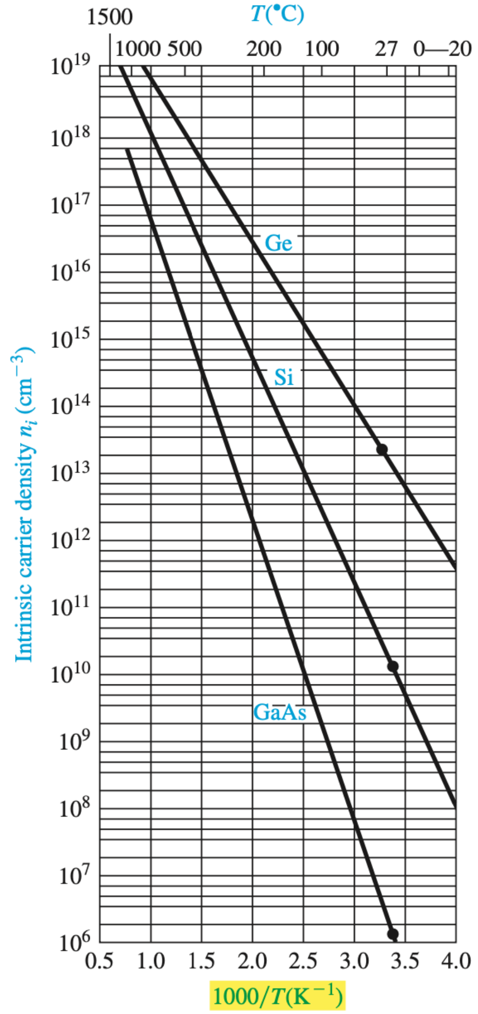


Figure 4.2 | The intrinsic carrier concentration of Ge, Si, and GaAs as a function of temperature.

n_i is very sensitive to the band gap value. Electron concertation is very different in intrinsic semiconductors

The Intrinsic Fermi-Level Position

Conditions for Intrinsic Semiconductor at Equilibrium

$n_i = p_i$

$$N_c \exp\left[\frac{-(E_c - E_{Fi})}{kT}\right] = N_v \exp\left[\frac{-(E_{Fi} - E_v)}{kT}\right] \tag{4.24}$$

If we take the natural log of both sides of this equation and solve for E_{Fi}, we obtain

$$E_{Fi} = \frac{1}{2}(E_c + E_v) + \frac{1}{2}kT \ln\left(\frac{N_v}{N_c}\right) \tag{4.25}$$

From the definitions for N_c and N_v given by Equations (4.10) and (4.18), respectively, Equation (4.25) may be written as

$$E_{Fi} = \frac{1}{2}(E_c + E_v) + \frac{3}{4}kT \ln\left(\frac{m_p^*}{m_n^*}\right) \tag{4.26a}$$

The first term, $\frac{1}{2}(E_c + E_v)$, is the energy exactly midway between E_c and E_v, or the midgap energy. We can define

$$\frac{1}{2}(E_c + E_v) = E_{\text{midgap}}$$

so that

$$\boxed{E_{Fi} - E_{\text{midgap}} = \frac{3}{4}kT \ln\left(\frac{m_p^*}{m_n^*}\right)} \tag{4.26b}$$

The intrinsic Fermi level must shift away from the band with the larger effective mass

also away from the larger density of states in order to maintain equal concentration of n and p.

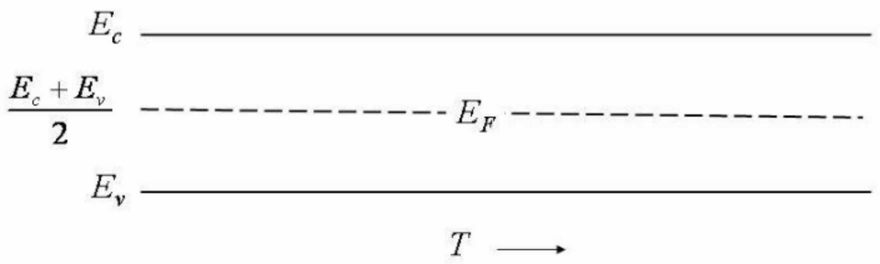
Intrinsic Semiconductors Band Gap and Fermi Energy

$$N_c\, e^{-(E_c - E_{Fi})/k_B T} = N_v\, e^{-(E_{Fi} - E_v)/k_B T}. \qquad E_{Fi} = \frac{E_c + E_v}{2} + \frac{k_B T}{2} \ln \frac{N_v}{N_c}.$$

$$\boxed{\text{If } m_n^* = m_p^*, \text{ then } N_v = N_c \text{ and } E_{Fi} = (E_c + E_v)/2 \text{ (midgap).}}$$

$$E_F = \frac{E_c + E_v}{2} + \frac{k_B T}{2} \ln\left(\frac{N_v}{N_c}\right)$$

Assumption
$m_n^* = m_p^*$



In Intrinsic semiconductors, with the assumption that effective masses are equal, then the Fermi Level is in the mid band gap

When calculating the actual Fermi level (e.g. non intrinsic case), the assumption of equal numbers of carriers may **no longer hold**, caused by an induced imbalance between electrons and holes

Example

Calculate the probability that a state in the conduction band is occupied by an electron and calculate the thermal equilibrium electron concentration in silicon ***(a)*** at T= 300 K , and ***(b)*** at T= 305 K.

Assume the Fermi energy is $0.25\, eV$ below the conduction band. The value of N_c for silicon at $T = 300\, K\ is\ Nc\ =\ 2.8\, x\, 10^{19} cm^{-3}$.

Solution

(a)

The probability that an energy state at $E\ = E_c$ is occupied by an electron is given by

$$n_{o1} = \ N_c \exp\left[\frac{-(E_c - E_F)}{kT}\right] = 2.8 \times 10^{19} \exp\left[\frac{-0.25}{8.62 \times 10^{-5} \times 300}\right]$$

$$n_o \ = \ 1.8\ \times 10^{15}\ cm^{-3}$$

(b)

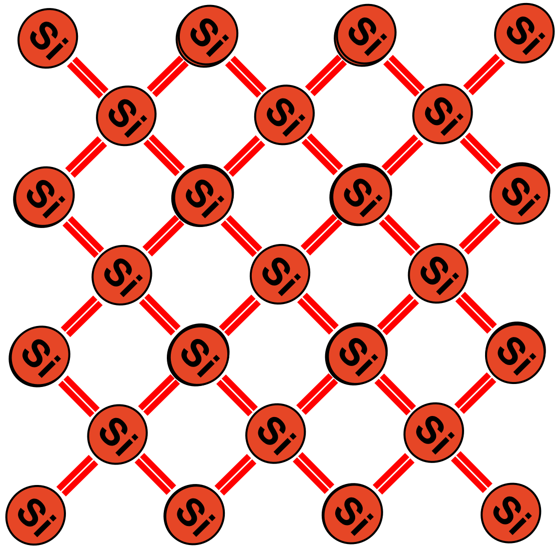
$$N_{c(2)} = N_{c(1)}\ \left[\frac{T_2}{T_1}\right]^{\frac{3}{2}} = 2.8 \times 10^{19} \left[\frac{305}{300}\right]^{\frac{3}{2}} = 2.87 \times 10^{19} cm^{-3}$$

$$n_{o2} = \ N_c \exp\left[\frac{-(E_c - E_F)}{kT}\right] = 2.87 \times 10^{19} \exp\left[\frac{-0.25}{8.62 \times 10^{-5} \times 305}\right]$$

$$= 3.75 \times\ 10^{19}\ cm^{-3}$$

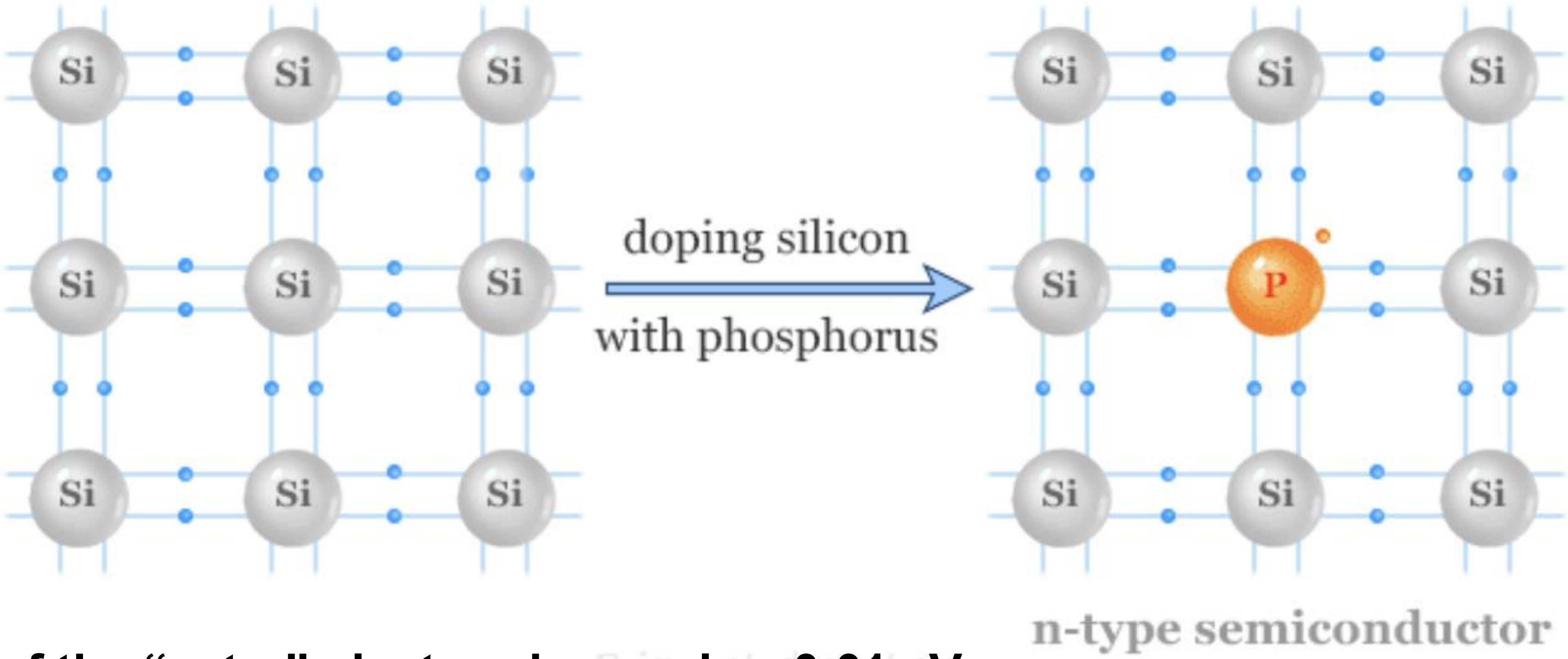
DOPANT ATOMS AND ENERGY LEVELS

- semiconductors
  - *Extrinsic*
    - n – type semiconductors
    - p – type semiconductors
  - Intrinsic
    - Pure semiconductors



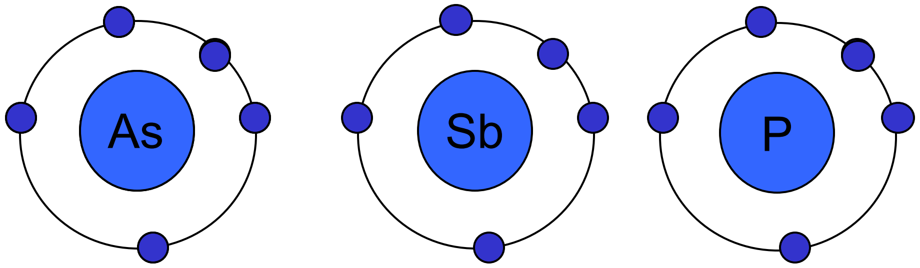
n type semiconductors

n – type semiconductor



Ionization Energy of the "extra" electron is very low 0.01 eV (kT is 0.0026 eV at room T)

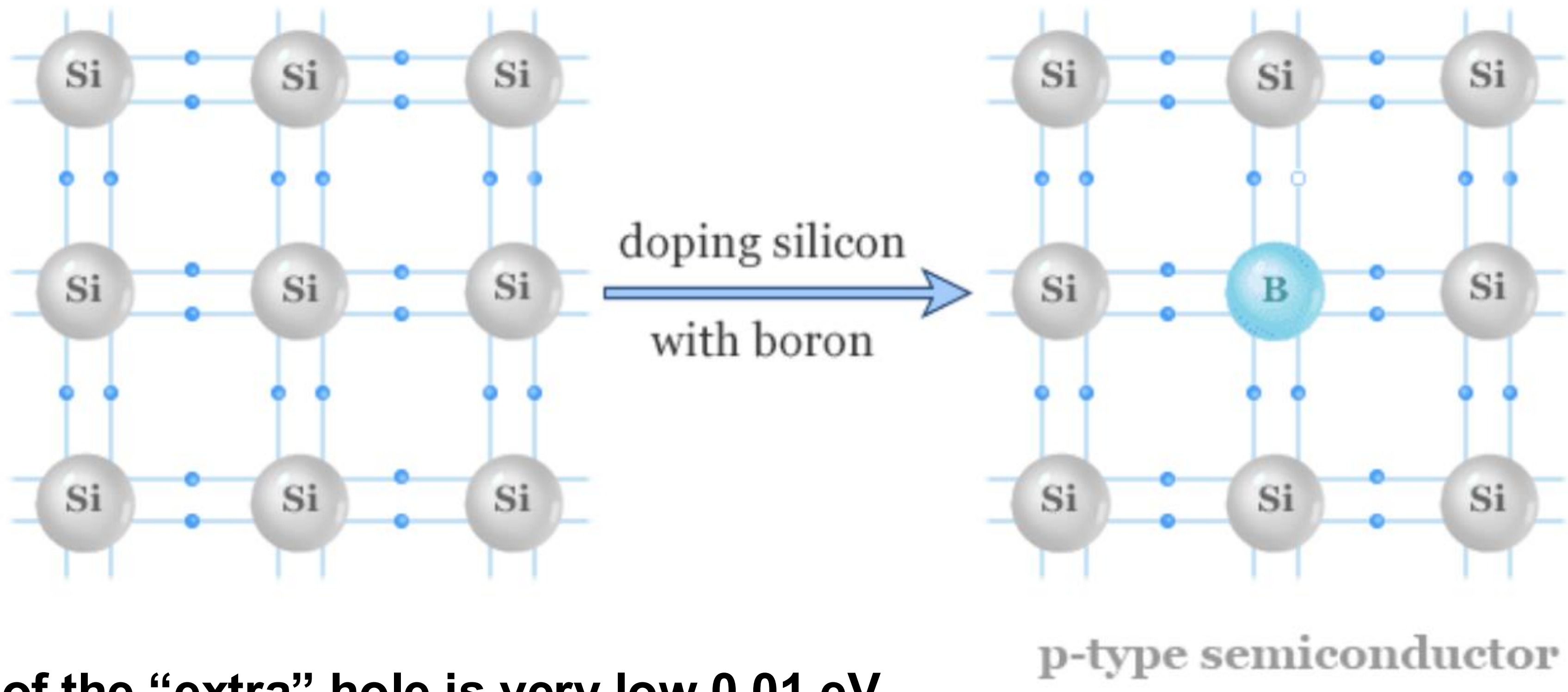
V equivalence



These atoms are called (donor atoms)

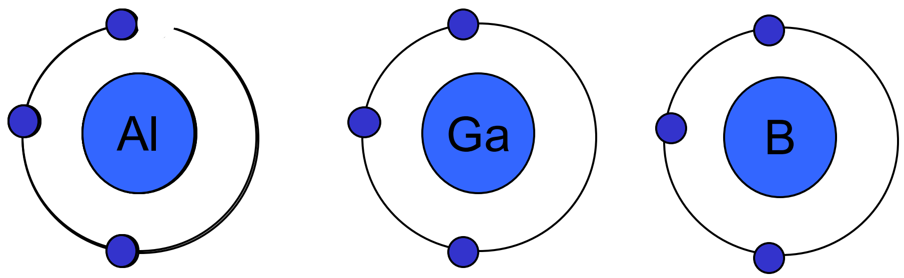
P type generated

p – type semiconductor



Ionization Energy of the “extra” hole is very low 0.01 eV
(kT is 0.0026 eV at room T)

III equivalence



These atoms are called
(acceptor atoms)

DOPING AND ENERGY LEVELS

- **Doping process:**
  - Add small amounts of impurity atoms (donors or acceptors).
  - Strongly modifies carrier concentrations while leaving lattice intact.

donor impurity atom
N-type semiconductor.

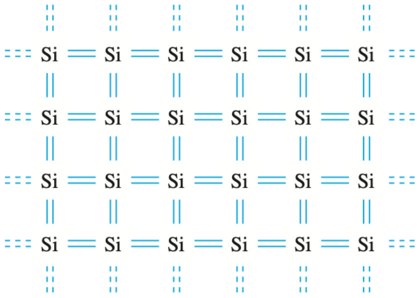


Figure 4.3 | Two-dimensional representation of the intrinsic silicon lattice.

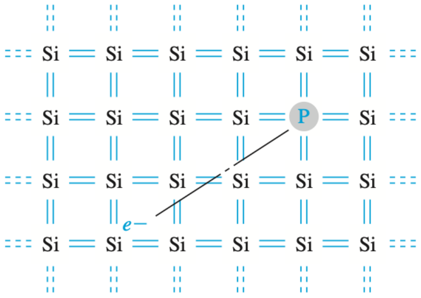


Figure 4.4 | Two-dimensional representation of the silicon lattice doped with a phosphorus atom.

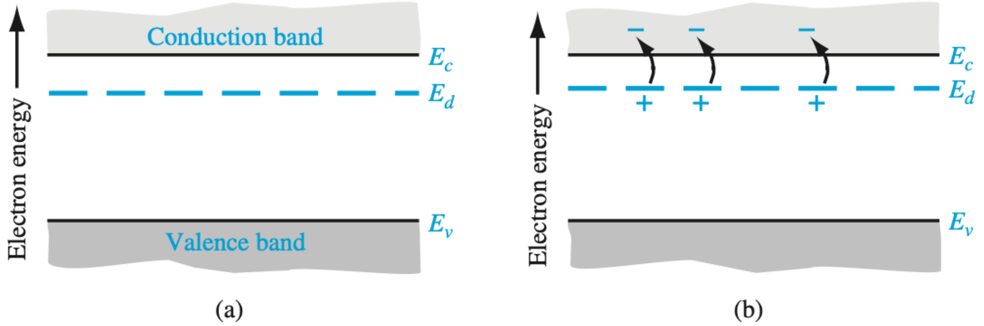


Figure 4.5 | The energy-band diagram showing (a) the discrete donor energy state and (b) the effect of a donor state being ionized.

The group V element has five valence electrons. Four of these will contribute to the covalent bonding with the silicon atoms, leaving the fifth more loosely bound to the dopant atom. We refer to the fifth valence electron as a donor electron.

DOPING AND ENERGY LEVELS

- **Doping process:**
  - Add small amounts of impurity atoms (donors or acceptors).
  - Strongly modifies carrier concentrations while leaving lattice intact.

acceptor impurity atom
P-type semiconductor.

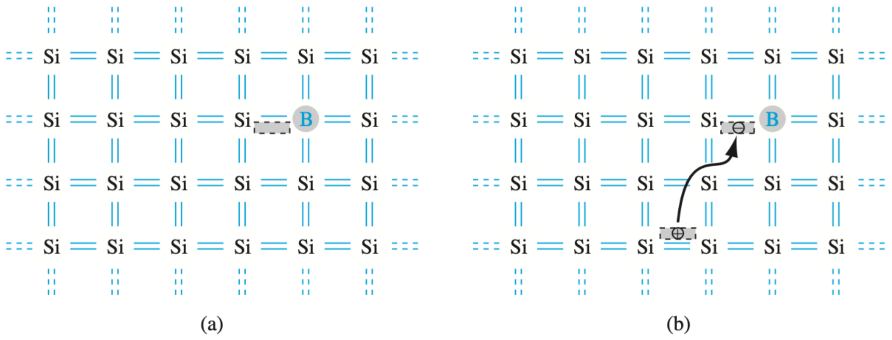


Figure 4.6 | Two-dimensional representation of a silicon lattice (a) doped with a boron atom and (b) showing the ionization of the boron atom resulting in a hole.

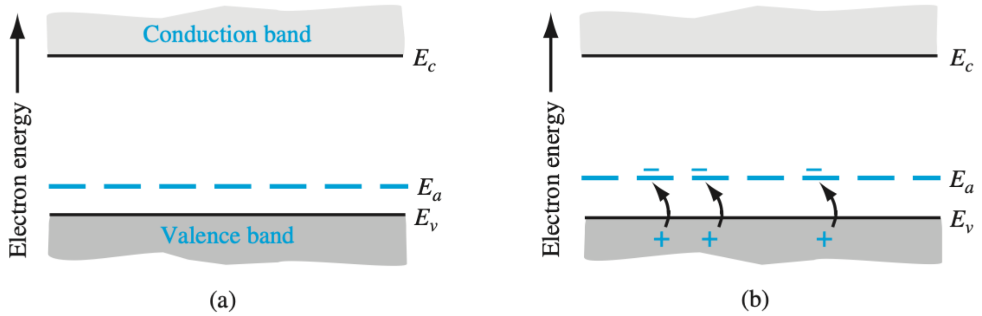


Figure 4.7 | The energy-band diagram showing (a) the discrete acceptor energy state and (b) the effect of an acceptor state being ionized.

The group III element has three valence electrons. These will each contribute to covalent bonding with the surrounding silicon atoms, leaving one bond incomplete, which can readily accept an electron from a neighbouring bond. We refer to this empty state as an acceptor level, and the dopant atom as an acceptor atom.

Ionization Energy

•A donor is like a "hydrogen atom inside the lattice".

•Magnitude of the Coulomb attraction and centripetal force

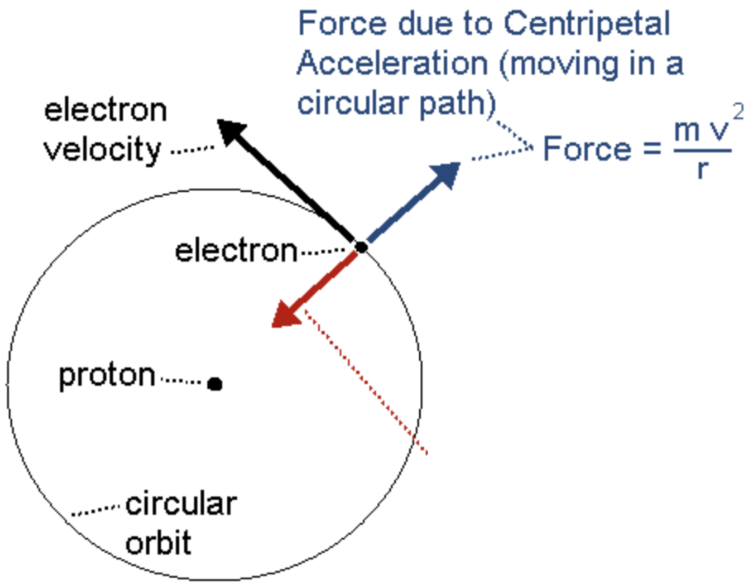
$$F_C = \frac{e^2}{4\pi\varepsilon\, r_n^2}, \qquad F_{\text{cent}} = \frac{m^* v^2}{r_n}, \qquad \longrightarrow \qquad \frac{e^2}{4\pi\varepsilon\, r_n^2} = \frac{m^* v^2}{r_n}$$

•Angular Momentum

$$m^* r_n v = n\hbar, \quad n = 1, 2, \ldots \qquad \longrightarrow \qquad v = \frac{n\hbar}{m^* r_n}.$$

•Combining the two relations: obtain the electron distance

$$\frac{e^2}{4\pi\varepsilon\, r_n^2} = \frac{m^*}{r_n}\left(\frac{n\hbar}{m^* r_n}\right)^2 = \frac{n^2\hbar^2}{m^* r_n^3}. \qquad \longrightarrow \qquad \boxed{r_n = \frac{(n\hbar)^2\, 4\pi\varepsilon}{m^* e^2}}$$



Ionization Energy

- **A donor is like a "hydrogen atom inside the lattice," with a weakly bound electron (binding energy ~ 10meV).**

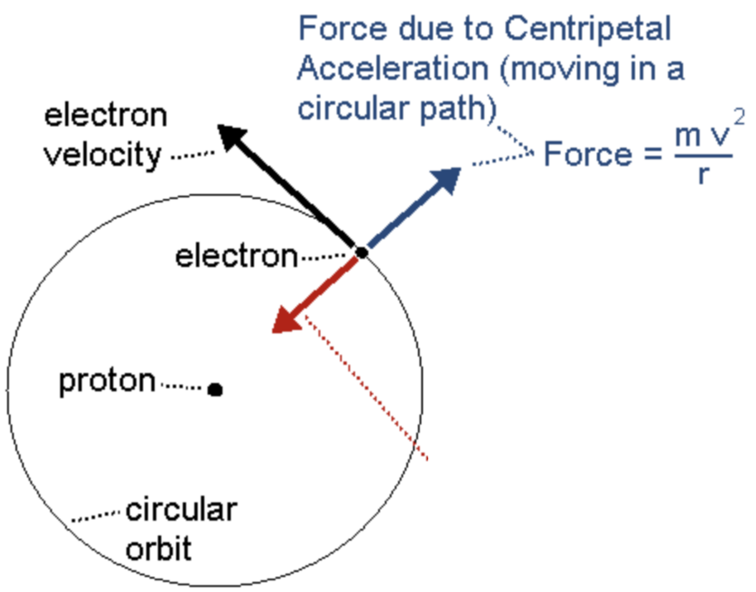
- Magnitude of the Coulomb attraction and centripetal force

$$E = T + V \quad (4.33), \qquad T = \frac{1}{2} m^* v^2 \quad (4.34), \qquad V(r) = -\frac{e^2}{4\pi\varepsilon\, r}.$$

- Calculating T and V using the quantized orbit

$$v = \frac{n\hbar}{m^* r_n}, \qquad r_n = \frac{4\pi\varepsilon\,\hbar^2}{m^* e^2} n^2 \implies T = \frac{1}{2} m^* \left(\frac{n\hbar}{m^* r_n}\right)^2 = \frac{m^* e^4}{2\,(n\hbar)^2\,(4\pi\varepsilon)^2}.$$

$$V = -\frac{e^2}{4\pi\varepsilon\, r_n} = -\frac{e^2}{4\pi\varepsilon} \cdot \frac{m^* e^2}{4\pi\varepsilon\,\hbar^2} \cdot \frac{1}{n^2} = -\frac{m^* e^4}{(n\hbar)^2\,(4\pi\varepsilon)^2}.$$



- Calculating the total energy

$$E = T + V = \frac{m^* e^4}{2\,(n\hbar)^2\,(4\pi\varepsilon)^2} - \frac{m^* e^4}{(n\hbar)^2\,(4\pi\varepsilon)^2} = -\frac{m^* e^4}{2\,(n\hbar)^2\,(4\pi\varepsilon)^2}.$$

$$\boxed{E_n = -\frac{m^* e^4}{2\,(n\hbar)^2\,(4\pi\varepsilon)^2}}$$

Ionization Energy

ionization energy: the approximate energy required to elevate the donor electron into the conduction band.

The analysis begins by setting the coulomb force of attraction between the electron and ion equal to the centripetal force of the orbiting electron. This condition we give a steady orbit. We have

$$\frac{e^2}{4\pi\epsilon {r_n}^2} = \frac{m^* v^2}{r_n}$$

where v is the magnitude of the velocity and r, is the radius of the orbit. If we assume the angular momentum is also quantized, then we can write

$$m^* r_n v = n\hbar$$

where n is a positive integer. Solving for v from Equation (4.28), substituting into Equation (4.27), and solving for the radius, we obtain

$$r_n = \frac{n^2 \hbar^2 4\pi \epsilon}{m^* e^2} \qquad (4.29)$$

Extrinsic Semiconductors

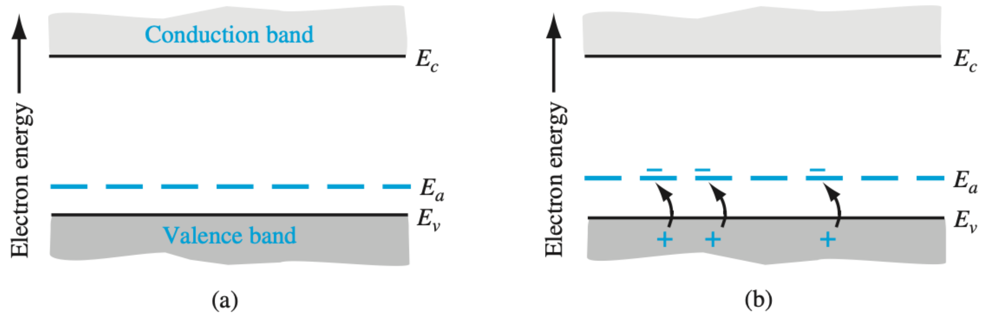


Figure 4.7 | The energy-band diagram showing (a) the discrete acceptor energy state and (b) the effect of an acceptor state being ionized.

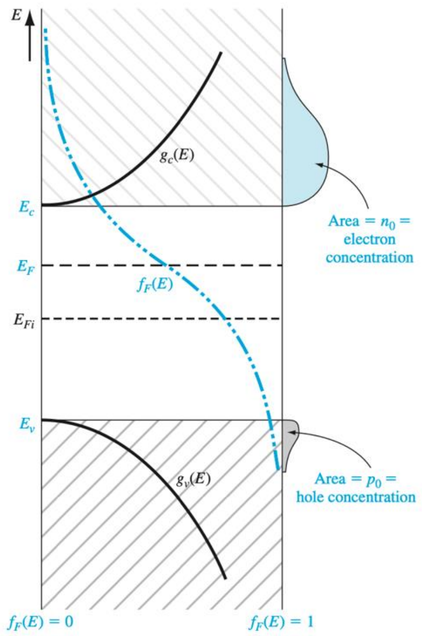


Figure 4.8 | Density of states functions, Fermi–Dirac probability function, and areas representing electron and hole concentrations for the case when E_F is above the intrinsic Fermi energy.

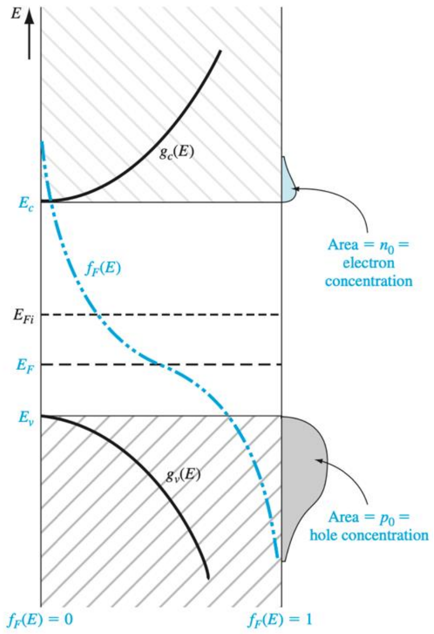


Figure 4.9 | Density of states functions, Fermi–Dirac probability function, and areas representing electron and hole concentrations for the case when E_F is below the intrinsic Fermi energy.

Majority and minority carrier

By adding dopants, the concentration of charge carriers (electrons and holes) is controlled, significantly enhancing the semiconductor's ability to conduct electricity.

Fermi Energy is shifted and concentration of electron and holes is not symmetric. E_f move close to VBM (p-type) or CBM (n-type)

Concentration of carrier loses symmetric distribution

POSITION OF FERMI ENERGY LEVEL

$$\boxed{E_c - E_F = kT \ln\left(\frac{N_c}{n_0}\right)} \tag{4.63}$$

$$\boxed{E_F - E_{Fi} = kT \ln\left(\frac{n_0}{n_i}\right)} \tag{4.65}$$

$$E_c - E_F = kT \ln\left(\frac{N_c}{N_d}\right) \tag{4.64}$$

$$\boxed{E_{Fi} - E_F = kT \ln\left(\frac{p_0}{n_i}\right)} \tag{4.68}$$

$$\boxed{E_F - E_v = kT \ln\left(\frac{N_v}{p_0}\right)} \tag{4.66}$$

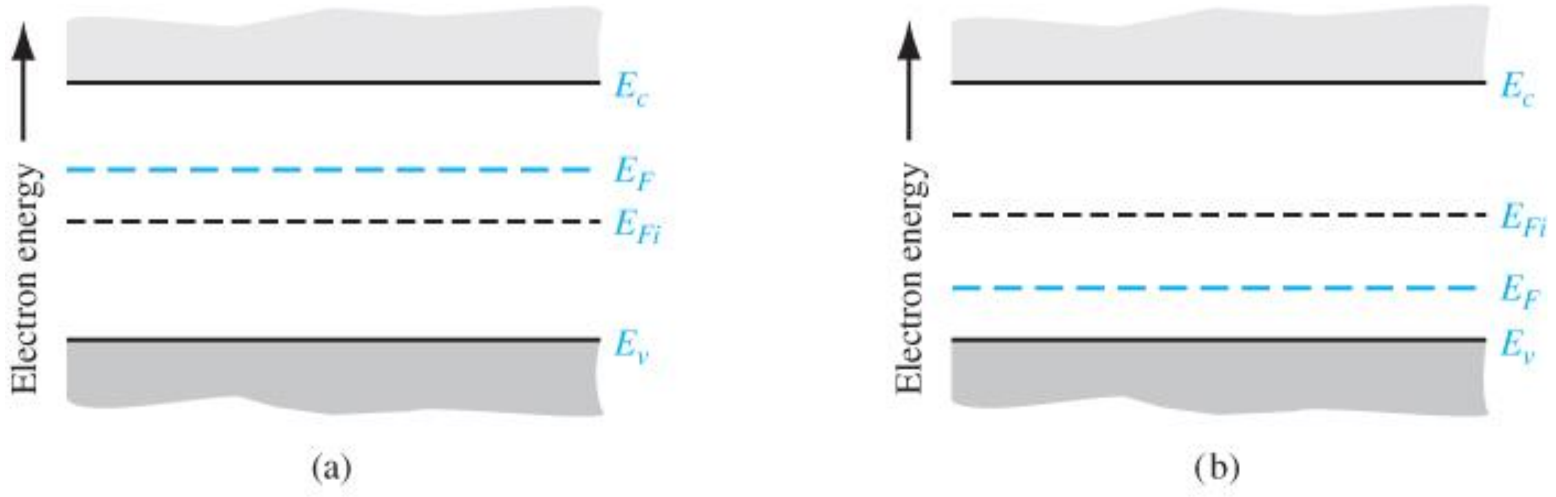


Figure 4.17 | Position of Fermi level for an (a) n-type ($N_d > N_a$) and (b) p-type ($N_a > N_d$) semiconductor.

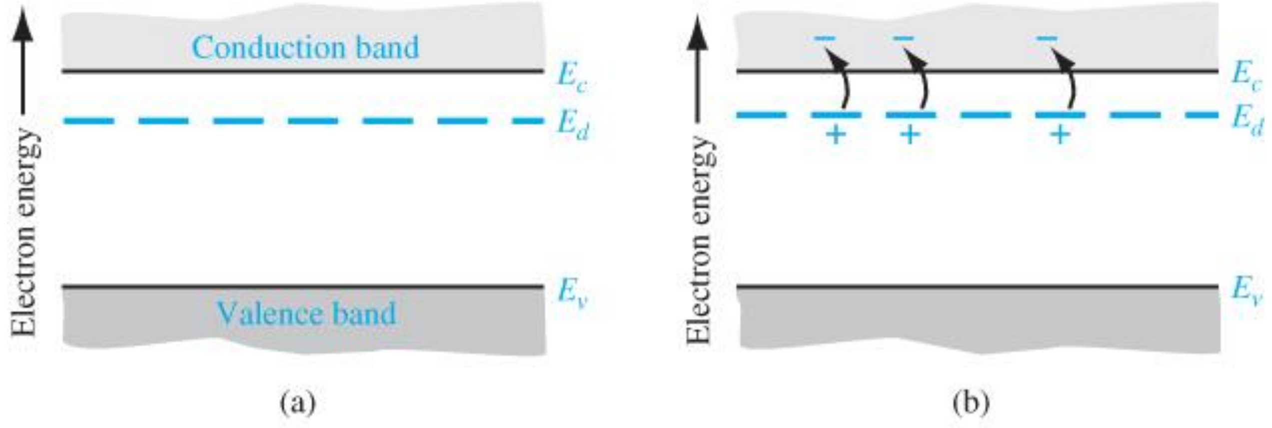


Figure 4.5 | The energy-band diagram showing (a) the discrete donor energy state and (b) the effect of a donor state being ionized.

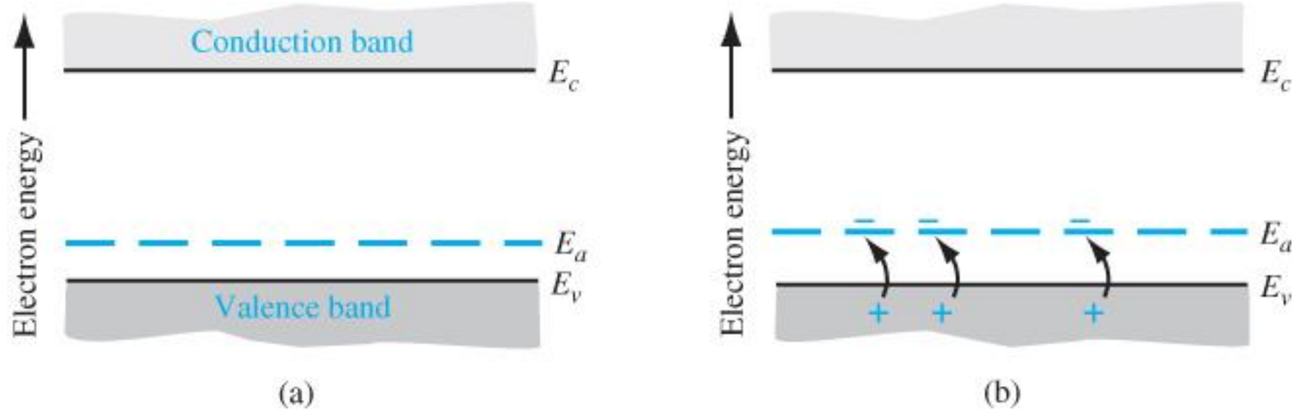


Figure 4.7 | The energy-band diagram showing (a) the discrete acceptor energy state and (b) the effect of an acceptor state being ionized.

Donor and acceptor atoms act as a *perturbation* to the perfect periodic potential of the crystal.

defects locally break the periodic boundary conditions assumed in the band structure model.

As a result, the electronic states they introduce are ***highly localised in real space* and do not have a well-defined k-space dependence**, unlike the delocalised Bloch states of the conduction or valence bands.

These electrons act not as free electrons, however the ionization energy is very small.

Ionization Energy

Ionization Energy = energy required to elevate the donor electron into the conduction band

Significance of Ionization Energy

•**Doping Efficiency:**

•The small ionization energy of dopants is crucial for creating extrinsic semiconductors. It ensures that a significant fraction of donor or acceptor atoms are ionized at room temperature, producing the free charge carriers (electrons or holes) needed for conduction.

•**Temperature Dependence:**

•The number of ionized dopants, and thus the conductivity, is temperature-dependent. As temperature increases, more thermal energy is available to ionize the dopants.

•**Contrast with Bandgap:**

•The ionization energy of dopants is orders of magnitude smaller than the semiconductor's bandgap energy. For instance, in silicon, the ionization energy for donors or acceptors is in the range of tens of millielectronvolts (meV), while the bandgap is around 1.12 electronvolts (eV) at room temperature

If we consider the lowest energy state in which $n = 1$, and if we consider silicon in which $\epsilon_r = 11.7$ and the conductivity effective mass is $m^*/m_0 = 0.26$, then we have that

$$\frac{r_1}{a_0} = 45 \tag{4.32}$$

or $r_1 = 23.9$ Å. This radius corresponds to approximately four lattice constants of

The donor electron is not tightly bound to the donor atom.

kinetic energy becomes

$$T = \frac{m^* e^4}{2(n\hbar)^2(4\pi\epsilon)^2} \tag{4.35}$$

The potential energy is

$$V = \frac{-e^2}{4\pi\epsilon r_n} = \frac{-m^* e^4}{(n\hbar)^2(4\pi\epsilon)^2} \tag{4.36}$$

The total energy is the sum of the kinetic and potential energies, so that

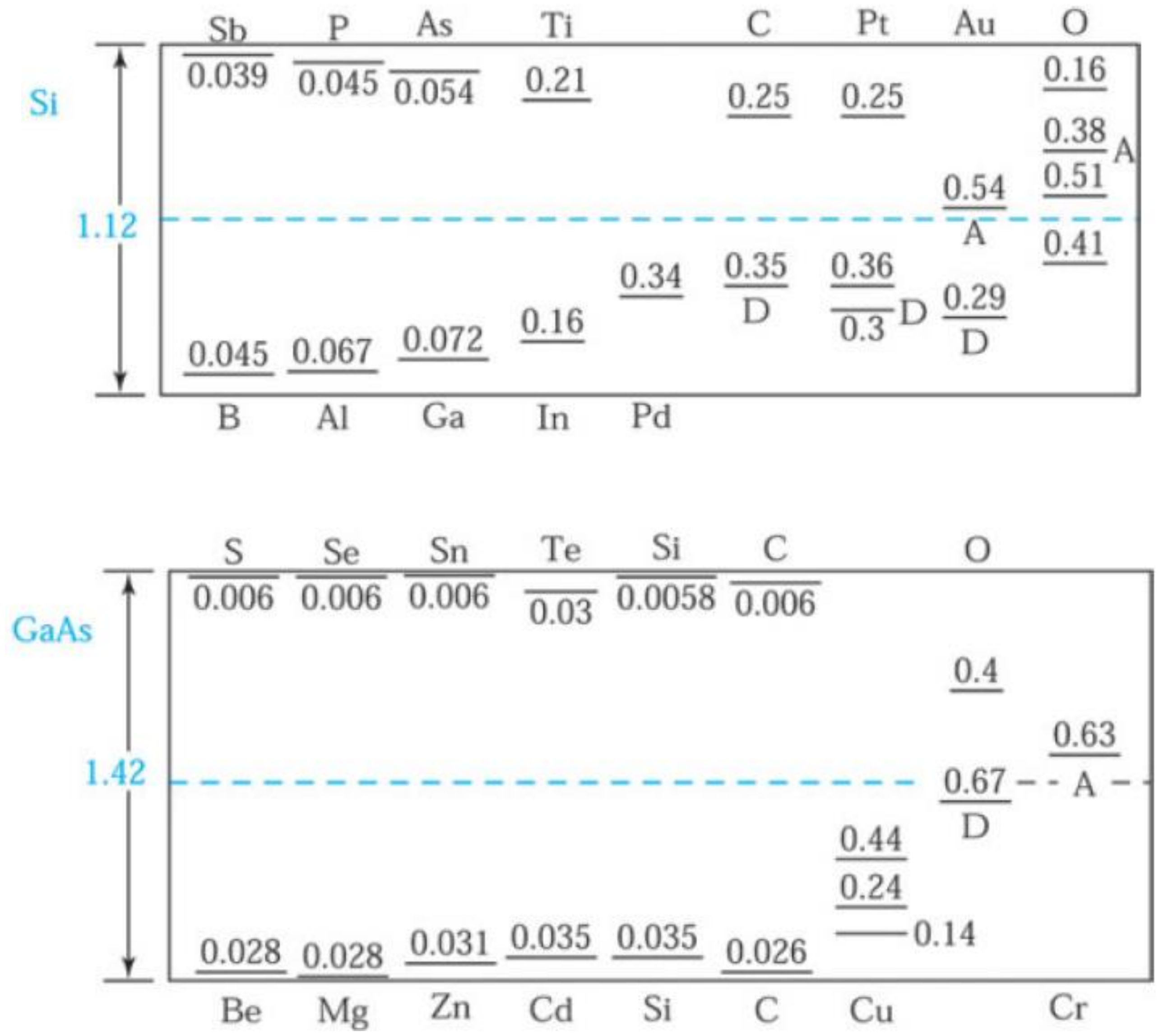
$$E = T + V = \frac{-m^* e^4}{2(n\hbar)^2(4\pi\epsilon)^2} \tag{4.37}$$

For silicon, the ionization energy is E = -25.8 meV, much less than the bandgap energy .

Table 4.3 | Impurity ionization energies in silicon and germanium

	Ionization energy (eV)	
Impurity	**Si**	**Ge**
Donors		
Phosphorus	0.045	0.012
Arsenic	0.05	0.0127
Acceptors		
Boron	0.045	0.0104
Aluminum	0.06	0.0102

Ionization Energy



Group III–V Semiconductors

- Group IV elements, such as silicon and germanium, can also be impurity atoms in gallium arsenide. Such impurities are called **amphoteric**
- **germanium is predominantly an acceptor and silicon is predominantly a donor.**

Table 4.4 | Impurity ionization energies in gallium arsenide

Impurity	Ionization energy (eV)
Donors	
Selenium	0.0059
Tellurium	0.0058
Silicon	0.0058
Germanium	0.0061
Acceptors	
Beryllium	0.028
Zinc	0.0307
Cadmium	0.0347
Silicon	0.0345
Germanium	0.0404

THE EXTRINSIC SEMICONDUCTOR

An **extrinsic semiconductor** is defined as a semiconductor in which controlled amounts of specific dopant or impurity atoms have been added so that the thermal-equilibrium electron and hole concentrations are different from the intrinsic carrier concentration.

- **The change in the Fermi level** is actually a function of the donor or acceptor impurity concentrations that are added to the semiconductor.
- electron and hole concentrations **change by orders of magnitude** from the intrinsic carrier concentration as the Fermi energy changes by a few tenths of an electron-volt.

- **The change in the Fermi level induce difference in carrier concentration**

$$\boxed{n_0 = n_i \exp\left[\frac{E_F - E_{Fi}}{kT}\right]} \qquad (4.39)$$

Similarly, if we add and subtract an intrinsic Fermi energy in the exponent of Equation (4.19), we will obtain

$$\boxed{p_0 = n_i \exp\left[\frac{-(E_F - E_{Fi})}{kT}\right]} \qquad (4.40)$$

The $n_o\, p_o$ Product

$$n_0 p_0 = N_c N_v \exp\left[\frac{-(E_c - E_F)}{kT}\right] \exp\left[\frac{-(E_F - E_v)}{kT}\right] \qquad (4.41)$$

which may be written as

$$\boxed{n_0 p_0 = N_c N_v \exp\left[\frac{-E_g}{kT}\right]} \qquad (4.42)$$

Law of mass action

The law of mass action states that the **product of concentration of electrons** in the conduction band and the concentration of **holes** in the valence band is **constant** at a **fixed temperature and is independent of amount of donor and acceptor** impurity added.

$$\boxed{n_0 p_0 = n_i^2}$$

Constrain for Extrinsic Semiconductors

The Fermi–Dirac Integral

valid. If the Boltzmann approximation does not hold, the thermal equilibrium electron concentration is written from Equation (4.3) as

$$n_0 = \frac{4\pi}{h^3}(2m_n^*)^{3/2} \int_{E_c}^{\infty} \frac{(E - E_c)^{1/2}\, dE}{1 + \exp\left(\frac{E - E_F}{kT}\right)}$$

$$\eta = \frac{E - E_c}{kT} \qquad (4.45a)$$

$$\eta_F = \frac{E_F - E_c}{kT} \qquad (4.45b)$$

$$F_{1/2}(\eta_F) = \int_0^{\infty} \frac{\eta^{1/2}\, d\eta}{1 + \exp(\eta - \eta_F)}$$

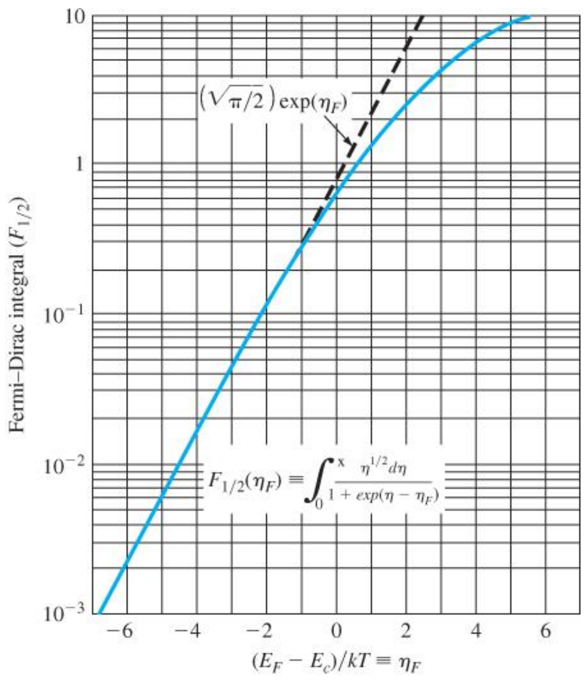


Figure 4.10 | The Fermi–Dirac integral $F_{1/2}$ as a function of the Fermi energy. (*From Sze [14].*)

When E_f is near or over E_C, and the Boltzmann approximation is not valid Fermi-Dirac integral expresses how filled the energy states are in the conduction band.

Degenerate and Nondegenerate Semiconductors

- As the donor concentration further increases, the band of donor states widens and may overlap the bottom of the conduction band.
- This overlap occurs when the donor concentration becomes comparable with the effective density of states.
- **When the concentration of electrons in the conduction band exceeds the density of states N_c**, the Fermi energy lies within the conduction band. This type of semiconductor is called a **degenerate** n-type semiconductor.
- In the degenerate n-type semiconductor, **the states between E_f and E_c are mostly filled with electrons**; thus, the electron concentration in the conduction band is very large.

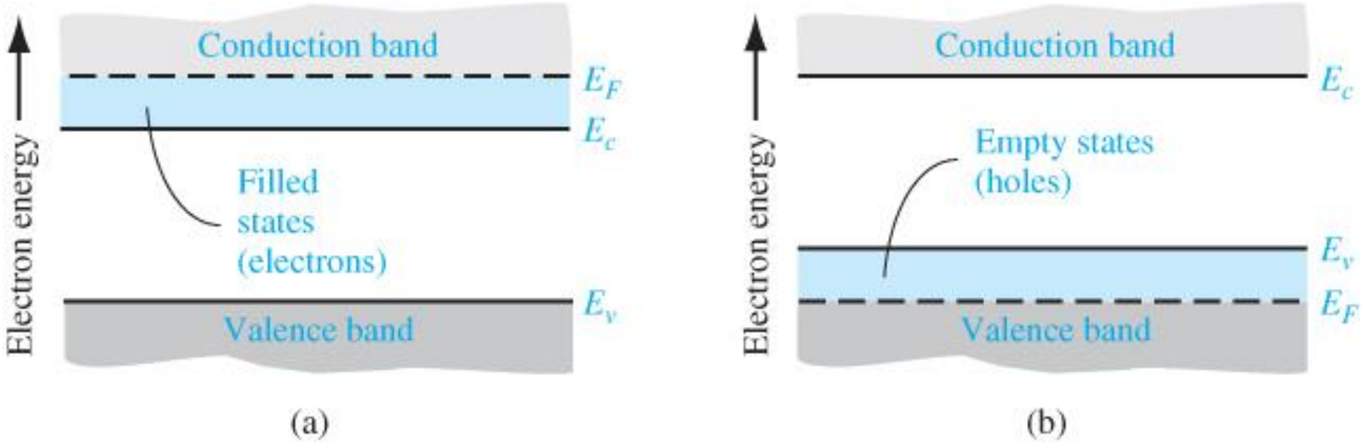


Figure 4.11 | Simplified energy-band diagrams for degenerately doped (a) n-type and (b) p-type semiconductors.

STATISTICS OF DONORS AND ACCEPTORS

Probability Function of donor and acceptor

The probability function of electrons occupying the donor state is

If Boltzmann approximation does not apply

$$n_d = \frac{N_d}{1 + \frac{1}{2}\exp\left(\frac{E_d - E_F}{kT}\right)} \tag{4.50}$$

where n_d is the concentration of electrons occupying the donor level and E_d is the energy of the donor level. The factor *1/2* in this equation is a direct result of the spin factor just mentioned.

$$\boxed{n_d = N_d - N_d^+} \tag{4.51}$$

where N_d^+ is the concentration of ionized donors. In many applications, we will be interested more in the concentration of ionized donors than in the concentration of electrons remaining in the donor states.

If we assume that $(E_d - E_F) \gg kT$, then

Complete Ionization

$$n_d \approx \frac{N_d}{\frac{1}{2}\exp\left(\frac{E_d - E_F}{kT}\right)} = 2N_d \exp\left[\frac{-(E_d - E_F)}{kT}\right] \quad (4.53)$$

$$\frac{n_d}{n_d + n_0} = \frac{1}{1 + \frac{N_c}{2N_d}\exp\left[\frac{-(E_c - E_d)}{kT}\right]} \quad (4.55)$$

The factor $(E_c - E_d)$ is just the ionization energy of the donor electrons.

there are very **few electrons in the donor state compared with the conduction band.** Essentially all of the electrons from the donor states are in the conduction band and, since only about 0.4 percent of the donor states contain electrons, **the donor states are said to be completely ionized**.

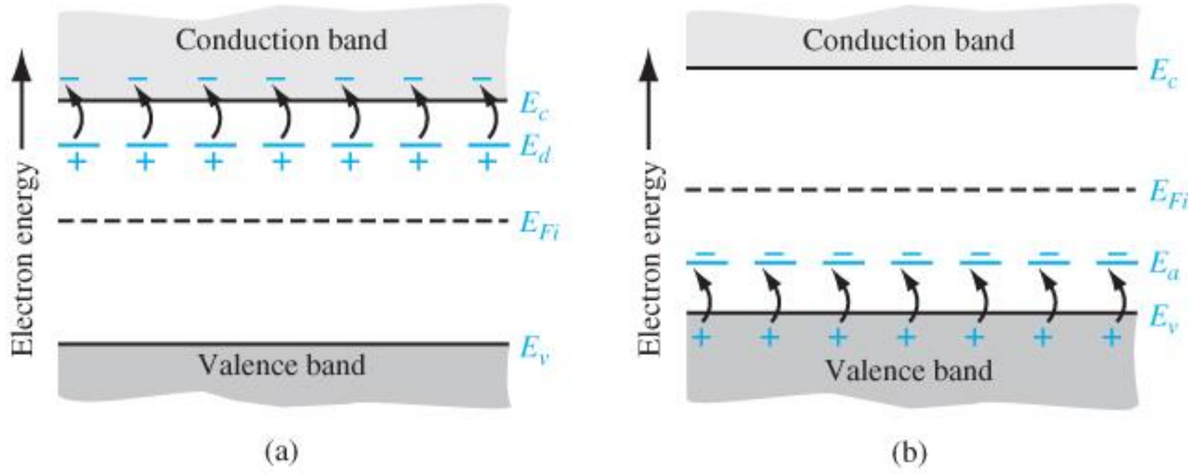


Figure 4.12 | Energy-band diagrams showing complete ionization of (a) donor states and (b) acceptor states.

Freeze-Out and Complete Ionization

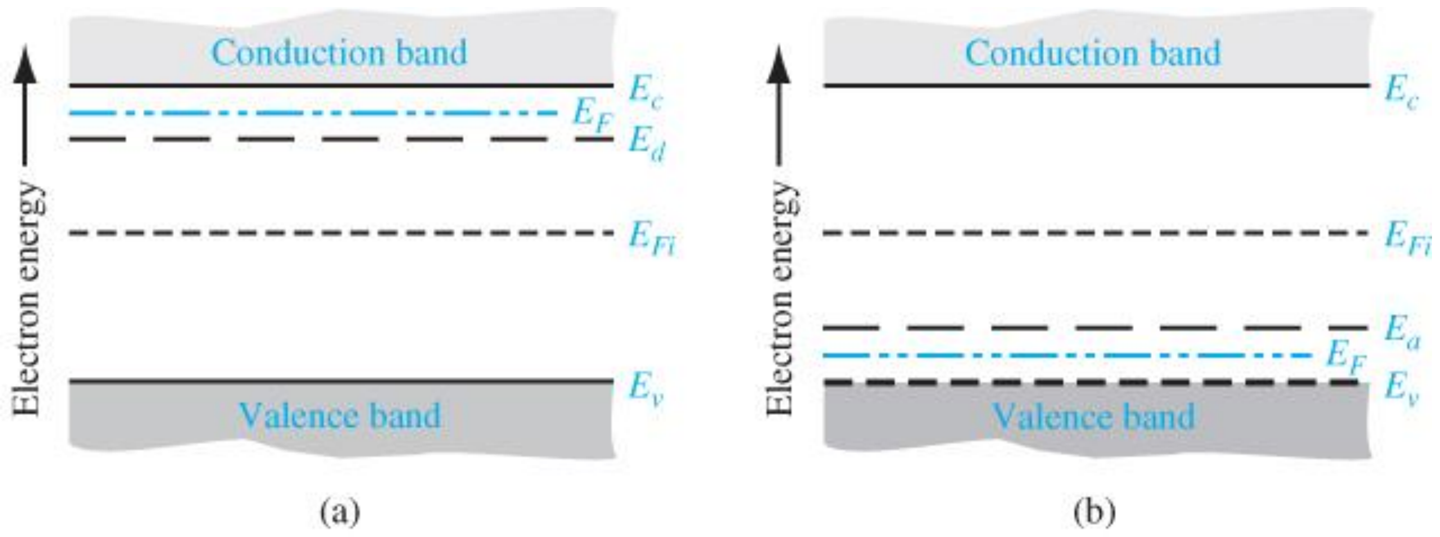


Figure 4.13 | Energy-band diagram at $T = 0$ K for (a) n-type and (b) p-type semiconductors.

at T = 0 K, No electrons from the donor state are thermally elevated into the conduction band; this effect is called freeze-out. Similarly, when no electrons from the valance band are elevated into the acceptor states, the effect is also called **freeze-out.**

at approximately 100C below room temperature, we still have 90 percent of the acceptor atoms ionized; in other words, 90 percent of the acceptor atoms have "donated" a hole to the valence band.

CHARGE NEUTRALITY

Compensated Semiconductors

A compensated semiconductor is one that contains both donor and acceptor impurity atoms in the same region.

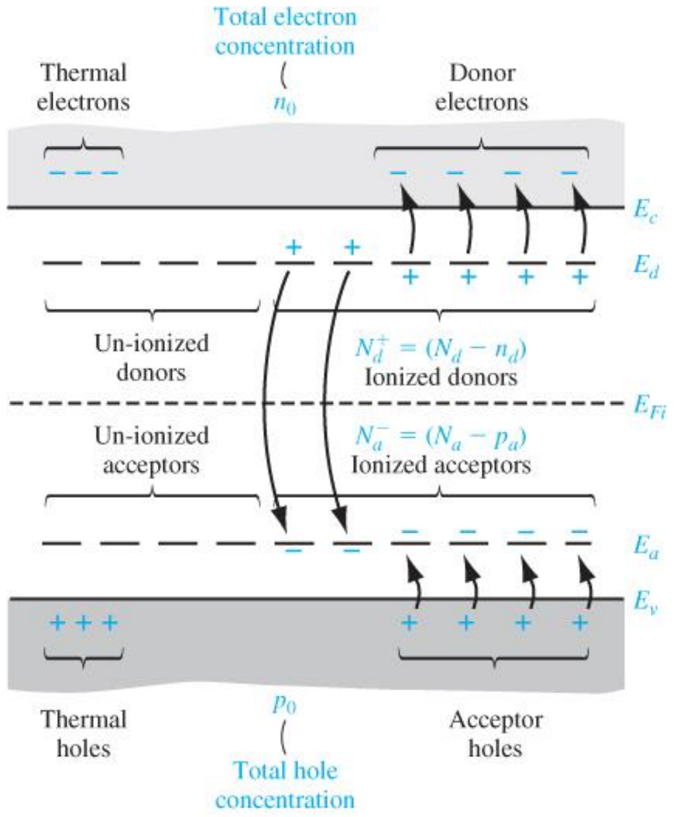


Figure 4.14 | Energy-band diagram of a compensated semiconductor showing ionized and un-ionized donors and acceptors.

Equilibrium Electron and Hole Concentrations

The charge neutrality condition is expressed by equating the density of negative charges to the density of positive charges.

$$n_0 + N_a^- = p_0 + N_d^+$$

- **Actual Fermi level** (in extrinsic semiconductors):
- Calculated using charge neutrality, which generally gives n ≠ p

Thermal-Equilibrium Electron Concentration If we assume complete ionization, n_d and p_a are both zero, and Equation (4.57) becomes

$$n_0 + N_a = p_0 + n_d \tag{4.58}$$

If we express p_0 as n_i^2/n_0, then Equation (4.58) can be written as

$$n_0 + N_a = \frac{n_i^2}{n_0} + N_d \tag{4.59a}$$

which in turn can be written as

$$n_0^2 - (N_d - N_a)n_0 - n_i^2 = 0 \tag{4.59b}$$

The electron concentration n_0 can be determined using the quadratic formula, or

$$\boxed{n_0 = \frac{(N_d - N_a)}{2} + \sqrt{\left(\frac{N_d - N_a}{2}\right)^2 + n_i^2}} \tag{4.60}$$

$(N_d - N_a) > n_i$, so the thermal-equilibrium majority carrier electron concentration is essentially equal to the difference between the donor and acceptor concentrations.

the majority carrier electron concentration is orders of magnitude larger than the minority carrier hole concentration.

A few of the donor electrons will fall into the empty states in the valence band and, in doing so, will **annihilate** some of the intrinsic holes.

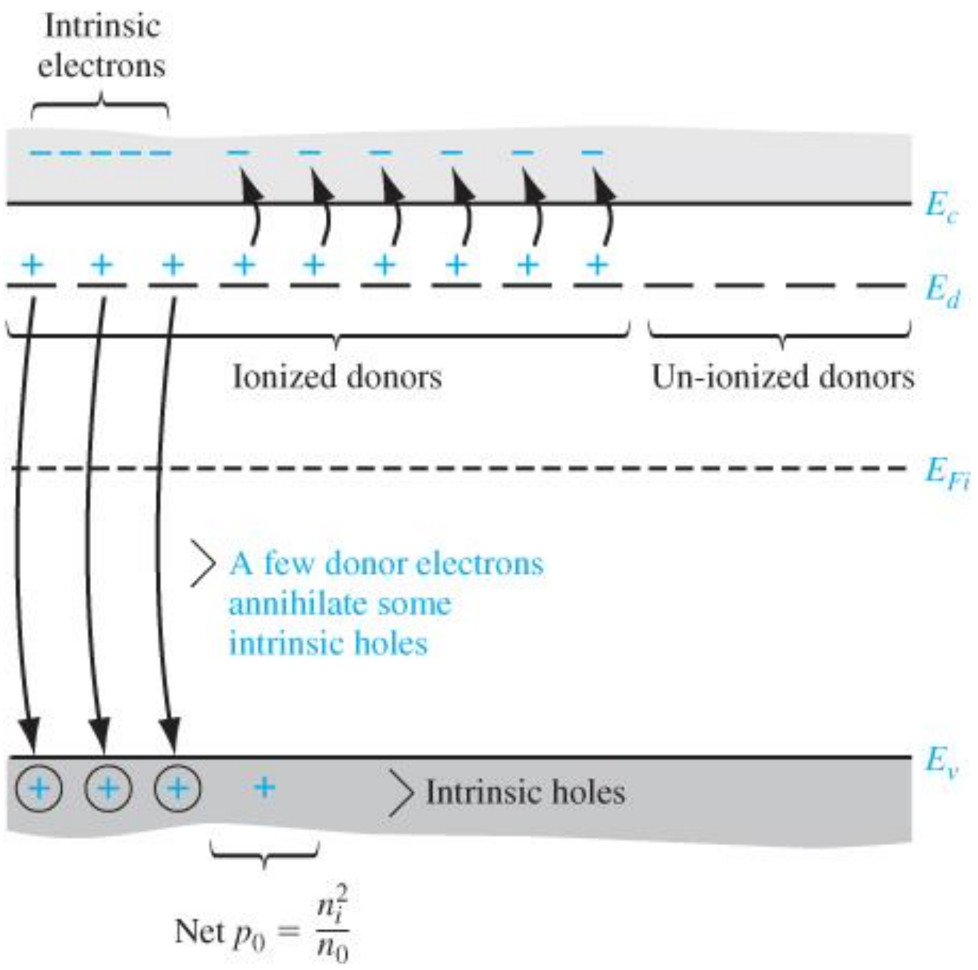


Figure 4.15 | Energy-band diagram showing the redistribution of electrons when donors are added.

If the **donor impurity concentration** **is not too different** in magnitude from the **intrinsic carrier concentration**, then the thermal-equilibrium majority carrier electron concentration is influenced by the intrinsic concentration.

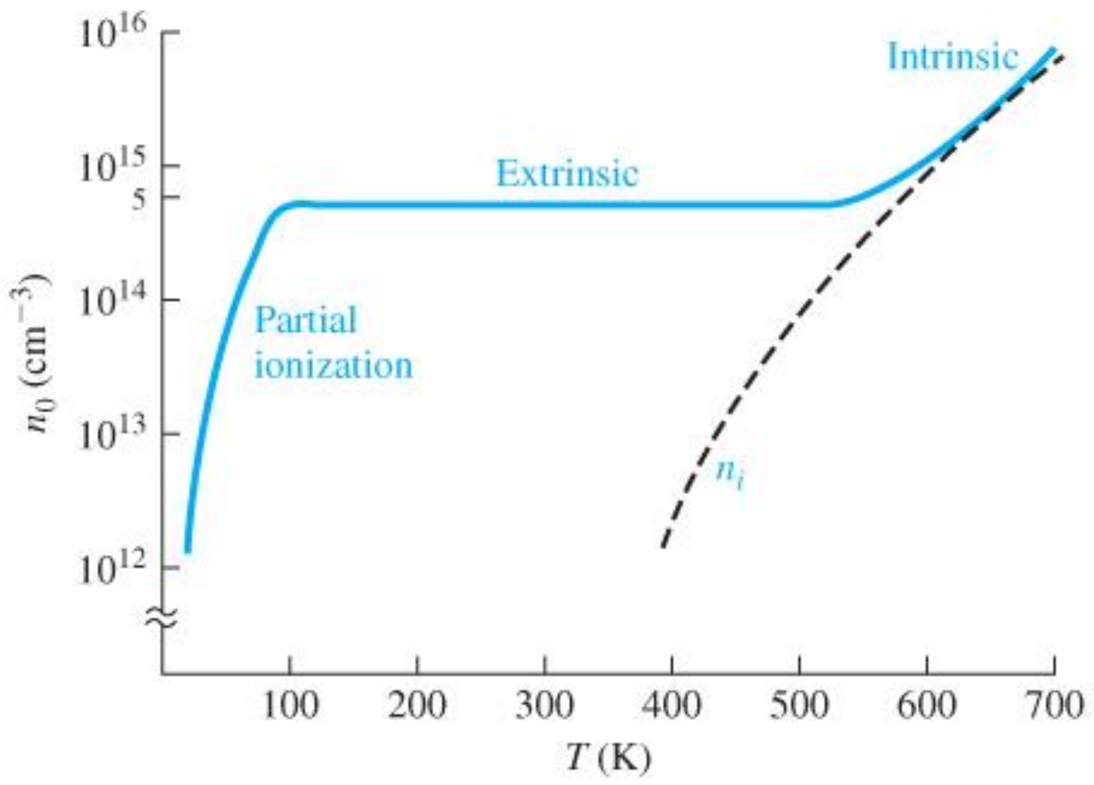


Figure 4.16 | Electron concentration versus temperature showing the three regions: partial ionization, extrinsic, and intrinsic.

Regime	**Majority Carrier (n-type)**	**Minority Carrier**
Freeze-out	$n \approx N_D e^{-(E_D - E_F)/k_B T} \ll N_D$	$p = \dfrac{n_i^2}{n}$
Extrinsic	$n \approx N_D$	$p = \dfrac{n_i^2}{N_D}$
Intrinsic	$n = n_i$	$p = n_i$

As the temperature increases, we can see where the intrinsic concentration begins to dominate. Also shown is the partial ionization, or the onset of freeze-out, at the low temperature.

Thermal-Equilibrium Hole Concentration If we reconsider Equation (4.58) and express n_0 as n_i^2/p_0, then we have

$$\frac{n_i^2}{p_0} + N_a = p_0 + N_d \tag{4.61a}$$

$$\boxed{p_0 = \frac{N_a - N_d}{2} + \sqrt{\left(\frac{N_a - N_d}{2}\right)^2 + n_i^2}} \tag{4.62}$$

If we assume complete ionization and if (Na - Nd) > ni, then the majority carrier hole concentration is, to a very good approximation, just **the difference between the acceptor and donor** concentrations.

Variation of E_F with Doping Concentration and Temperature

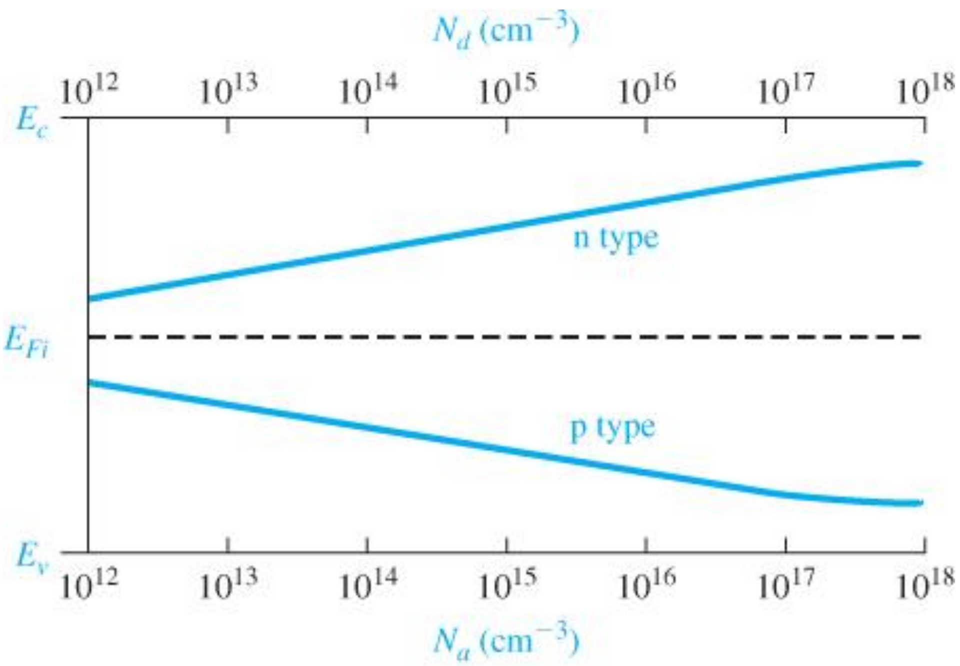


Figure 4.18 | Position of Fermi level as a function of donor concentration (n type) and acceptor concentration (p type).

If the acceptor (or donor) concentration in silicon is **greater than approximately 3 X 10^{17} cm^{-3}**, then the Boltzmann approximation of the distribution function **becomes less valid** and the equations for the Fermi-level position are no longer quite as accurate.

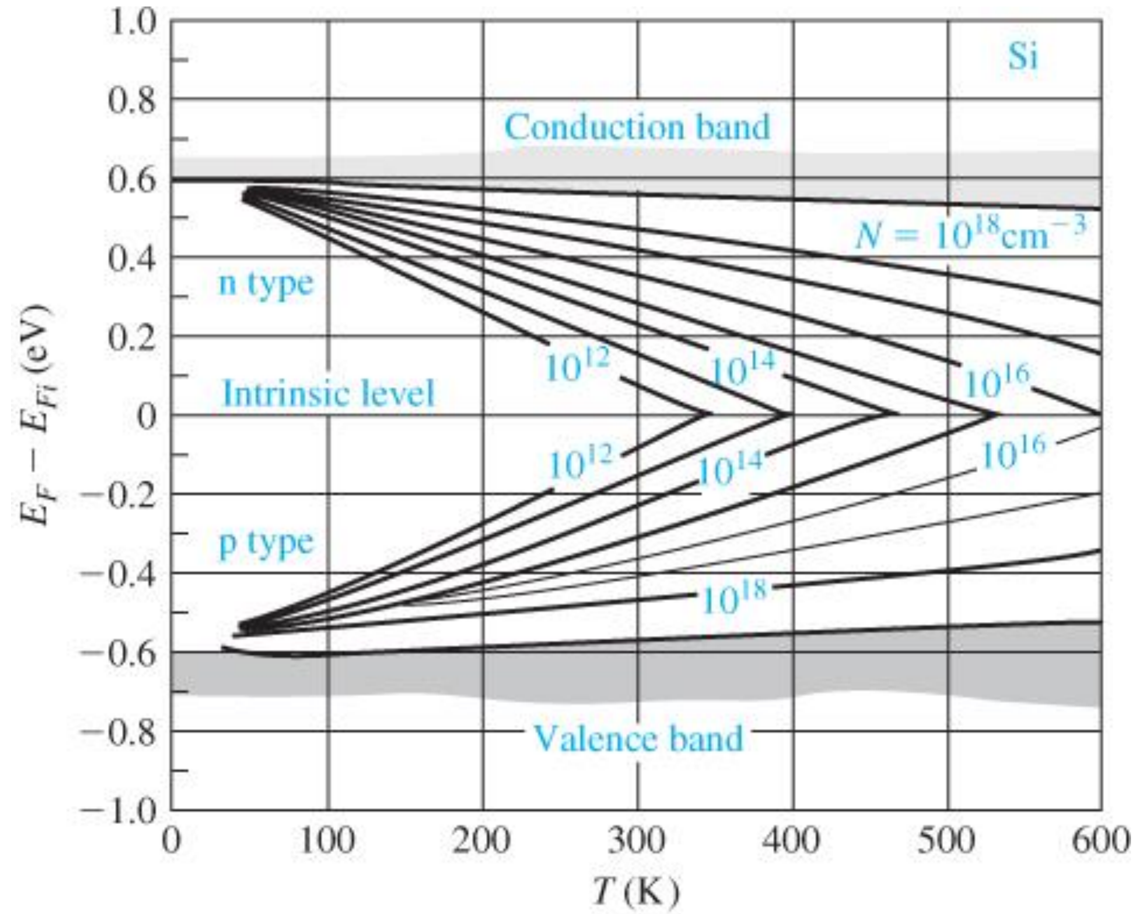


Figure 4.19 | Position of Fermi level as a function of temperature for various doping concentrations. *(From Sze [14].)*

At the low temperature where freeze-out occurs, the Fermi level goes above E_d for the n-type material and below E_a for the p-type material.

In thermal equilibrium, the Fermi energy level is a constant throughout a system.

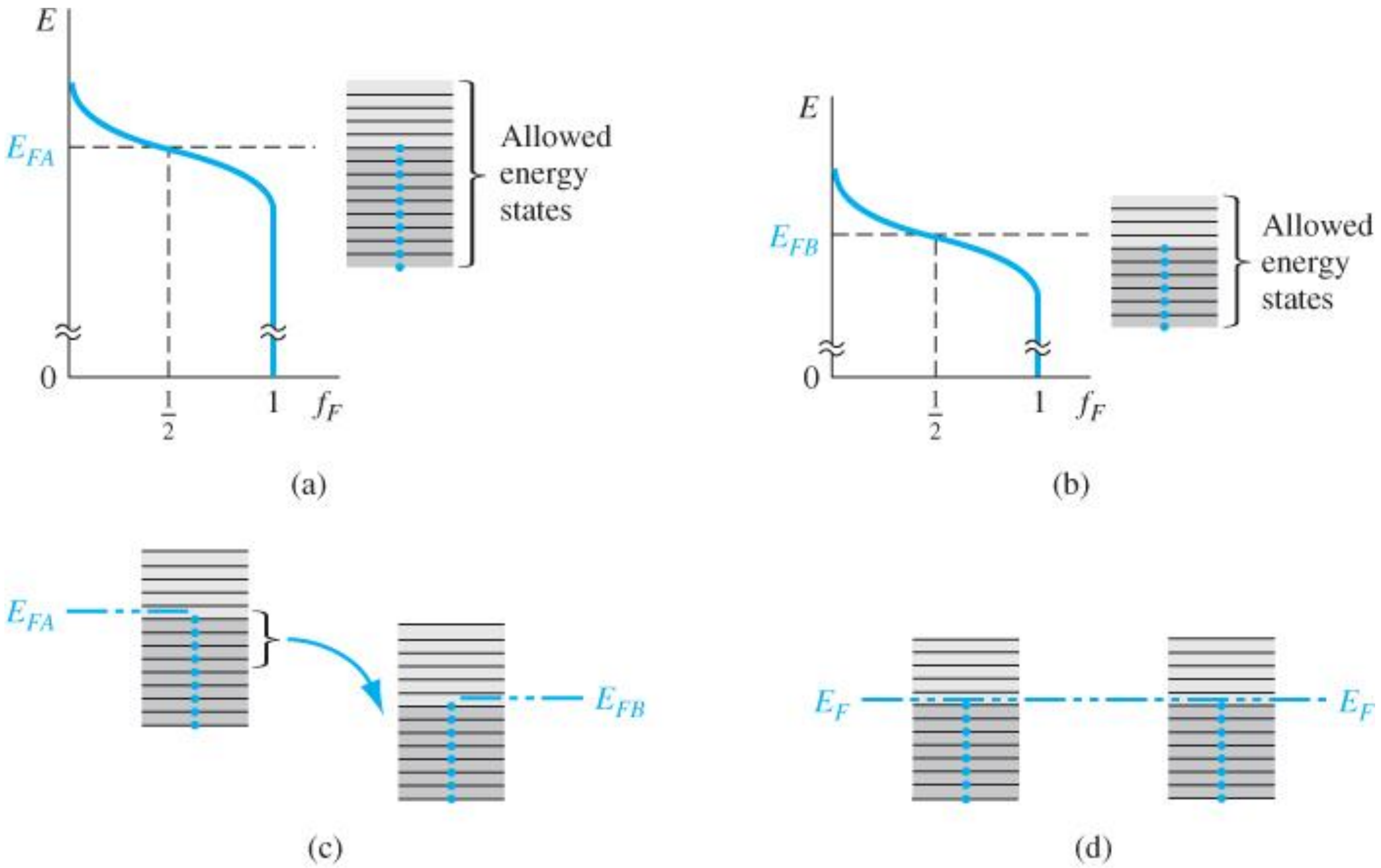


Figure 4.20 | The Fermi energy of (a) material A in thermal equilibrium, (b) material B in thermal equilibrium, (c) materials A and B at the instant they are placed in contact, and (d) materials A and B in contact at thermal equilibrium.